Tutorial - Week 4

TG 11 - TG 12

1. Explain why the function is discontinuous at the given number a.

(a) $f(x) = \frac{x+1}{x^2-1}$, $a = 1$ and $a = -1$.

(b) $f(x) = \tan x$, $a = \frac{\pi}{2}$.

(c) $f(x) = \begin{cases} e^x & \text{if } x < 1, \\ 3 & \text{if } x \geq 1. \end{cases}$ $a = 1$.

(d) $f(x) = \begin{cases} \sin(x) & \text{if } x \leq 0, \\ \ln x & \text{if } x > 0. \end{cases}$ $a = 0$.

(e) $f(x) = \begin{cases} \frac{x^2+x}{x^2-1} & \text{if } |x| \neq 1, \\ 0 & \text{if } |x| = 1. \end{cases}$ $|a| = 1$.

2. Use the definition of continuity and the properties of limits to show that the function is continuous on the given interval.

(a) $f(x) = x^2 - \sqrt{-x^2+9}$, $[-3, 3]$.

(b) $f(x) = \frac{x^3-x^2+x}{1-\sin(x)}$, $[-\frac{\pi}{2}, \frac{\pi}{2})$.

3. Find the numbers at which f is discontinuous. At which of these numbers is f continuous from the right, from the left, or neither?

(a) $f(x) = \begin{cases} x^3 & \text{if } x < -1, \\ x & \text{if } -1 \leq x \leq 1, \\ x^2 & \text{if } x \geq 1, \end{cases}$

(b) $f(x) = \begin{cases} -\sin x & \text{if } x < -\pi, \\ \cos\left(x - \frac{\pi}{2}\right) & \text{if } -\pi \leq x < \pi, \\ \tan x & \text{if } x \geq \pi, \end{cases}$

4. Find the values of the parameters that make f continuous everywhere.

(a) $f(x) = \begin{cases} \frac{2x^2-x+1}{x-1} & \text{if } x \leq 0, \\ ax+b & \text{if } 0 < x < 2, \\ e^x & \text{if } x \geq 2. \end{cases}$

(b) $f(x) = \begin{cases} ax^2 - b & \text{if } x \leq 0, \\ 2\cos x & \text{if } 0 < x \leq c, \\ \cos^2 x & \text{if } c < x \leq \pi, \\ ax^2 + b & \text{if } x > \pi. \end{cases}$

5. Let f be a continuous function at c with $f(c) \neq 0$. Prove that there exists an interval $(c-\delta, c+\delta)$ in which $f(c)$ and $f(x)$ have the same sign.

6. Prove that if n is a positive integer and $a > 0$, there exists a unique $b > 0$ such that $b^n = a$.

7. Prove that if $r > 0$ is a rational number, then $\lim_{x\to\infty} \frac{1}{x^r} = 0$.

8. Find the limit.

(a) $\lim_{x\to\infty} \frac{3x-1}{-3x+2}$.

(b) $\lim_{x\to-\infty} \frac{x+3}{x^2-9}$.

(c) $\lim_{x\to\infty} \left(\sqrt{x^2+ax} - \sqrt{x^2+bx}\right)$.

(d) $\lim_{x\to\infty} \left(\ln(x^2-1) - \ln(x^2+2)\right)$.

(e) $\lim_{x\to-\infty} e^{3x}\sin x$.

(f) $\lim_{x\to-\infty} \frac{\sqrt{1+4x^6}}{2-x^3}$.

9. Find the horizontal and vertical asymptotes of each curve.

(a) $f(x) = \frac{x-1}{x-2}$.

(b) $f(x) = \frac{2x-3}{x+1}$.

(c) $f(x) = \frac{x^2-1}{x^2-2}$.

(d) $f(x) = \frac{2e^x-3}{3e^x-2}$.

(e) $f(x) = \ln(x+1) - \ln x$.

(f) $f(x) = \ln x - \ln(1-x)$.

10. Consider $p(x)$ and $q(x)$ two polynomials, find $\lim_{x\to\infty} \dfrac{p(x)}{q(x)}$.

11. Prove that $\lim_{x\to\infty} f(x) = \lim_{x\to 0^+} f\left(\dfrac{1}{x}\right)$, if the limit exists.

12. Find the derivatives of the following functions by applying the definition of derivative as a limit.

(a) $f(x) = x^4$.

(b) $f(x) = \sin x$.

(c) $f(x) = \frac{1}{x}$.

(d) $f(x) = \frac{1}{\sqrt{x}}$.

13. For the following values of f and a, find the equation of the tangent line to the graph of f at the point $(a, f(a))$.

(a) $f(x) = x^4,\ a = 2$.

(b) $f(x) = \sin x,\ a = \frac{\pi}{2}$.

(c) $f(x) = \frac{1}{x},\ a = -2$.

(d) $f(x) = \frac{1}{\sqrt{x}},\ a = 3$.

14. (a) Let f be a function such that $|f(x)| \leq x^2$ for all x on some interval $(-a, a)$. Prove that f is differentiable at 0.

(b) Let f be the function defined by

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Where is f differentiable?

15. Find a, b, c and d in $\mathbb{R}$ such that the following function is differentiable in $\mathbb{R}$:

$$f(x) = \begin{cases} \sin(2x) & \text{if} \quad x \leq 0, \\ ax^3 + bx^2 + cx + d & \text{if} \quad 0 < x < 1, \\ \dfrac{1}{x} & \text{if} \quad x \geq 1. \end{cases}$$

For the values you found, is f' continuous? Is it differentiable?